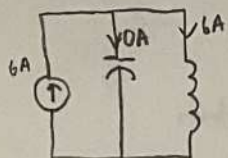


a) $t = (0^-)$ 

$$i_L(0^-) = i_L(0^+) = 6A$$

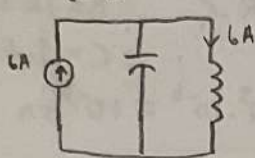
$$V_C(0^-) = V_C(0^+) = 0$$

$$i_R(0^-) = i_R(0^+) = 0$$

$$V_L(0^-) = V_L(0^+) = 0$$

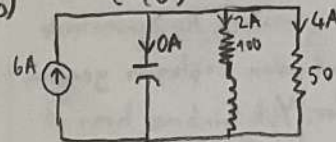
$$i_C(0^-) = i_C(0^+) = 0$$

$$\frac{dV_C(0^+)}{dt} = 0 \text{ V/s}$$

 $t = 0^+$ 

$$i_C = C \frac{dV_C}{dt}$$

$$\frac{dV_C}{dt} = \frac{i_C}{C} = \frac{0}{0,01} = 0$$

b) $t = (0^-)$ 

$$i_L(0^-) = i_L(0^+) = 2A$$

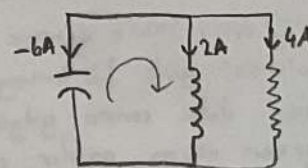
$$V_C(0^-) = V_C(0^+) = 200V$$

$$i_R(0^-) = i_R(0^+) = 4A$$

$$V_L(0^+) = 200V$$

$$i_C(0^+) = -6A$$

$$\frac{dV_C(0^+)}{dt} = -600 \text{ V/s}$$

 $t = (0^+)$ 

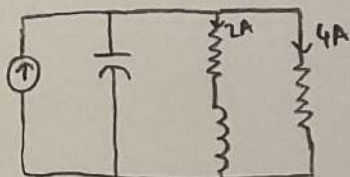
$$4 \cdot 50 \Omega = 200V$$

$$-V_C + V_L = 0$$

$$V_C = V_L$$

$$V_C = 200$$

$$\frac{dV_C}{dt} = \frac{i_C}{C} = \frac{-6}{0,01} = -600$$

c) $t = (0^-)$ 

$$i_L(0^-) = i_L(0^+) = 2A$$

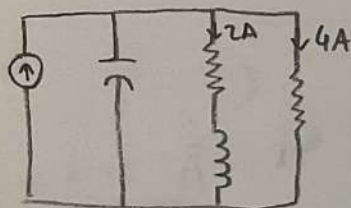
$$V_C(0^-) = V_C(0^+) = 200V$$

$$i_R(0^-) = i_R(0^+) = 4A$$

$$V_L(0^-) = V_L(0^+) = 0V$$

$$i_C(0^-) = i_C(0^+) = 0A$$

$$\frac{dV_C(0^+)}{dt} = 0 \text{ V/s}$$

 $t = (0^+)$ 

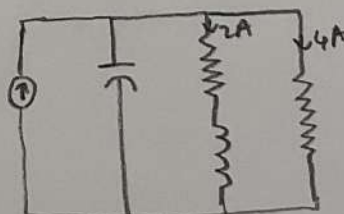
$$-V_C + 2 \cdot 100 \Omega + V_C = 0$$

$$V_L = V_C - 200$$

$$V_L = 200 - 200$$

$$V_L = 0$$

$$\frac{i_C}{C} = \frac{0}{0,01} = 0$$

d) $t = (0^-)$ 

$$i_L(0^-) = i_L(0^+) = 2A$$

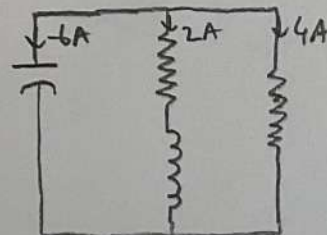
$$V_C(0^-) = V_C(0^+) = 200V$$

$$i_R(0^-) = i_R(0^+) = 4A$$

$$V_L(0^-) = V_L(0^+) = 0V$$

$$i_C(0^+) = -6A$$

$$\frac{dV_C(0^+)}{dt} = -600 \text{ V/s}$$

 $t = (0^+)$ 

$$\frac{i_C}{C} = \frac{-6}{0,01} = -600 \text{ V/s}$$